

REST CRUD Lab (CodeGrade)

 <https://github.com/learn-co-curriculum/python-p4-flask-restful-crud-lab>  <https://github.com/learn-co-curriculum/python-p4-flask-restful-crud-lab/issues/new>

Learning Goals

- Build RESTful APIs that are easy to navigate and use in applications.
- Develop a Flask API with successful frontend connections via `fetch()`.
- Integrate update and delete routes with the associated actions to return the appropriate JSON data.

Key Vocab

- **Representational State Transfer (REST)**: a convention for developing applications that use HTTP in a consistent, human-readable, machine-readable way.
- **Application Programming Interface (API)**: a software application that allows two or more software applications to communicate with one another. Can be standalone or incorporated into a larger product.
- **HTTP Request Method**: assets of HTTP requests that tell the server which actions the client is attempting to perform on the located resource.
- **GET** : the most common HTTP request method. Signifies that the client is attempting to view the located resource.
- **POST** : the second most common HTTP request method. Signifies that the client is attempting to submit a form to create a new resource.
- **PATCH** : an HTTP request method that signifies that the client is attempting to update a resource with new information.
- **PUT** : an HTTP request method that signifies that the client is attempting to update a resource with new information contained in a complete record.
- **DELETE** : an HTTP request method that signifies that the client is attempting to delete a resource.

Introduction

In this lab, we'll continue building an API for the plant store! The code for the frontend React application is done; you can find it in the `client` directory. Your job is to create the Flask API so that the `fetch` requests on the frontend work successfully.

Instructions

To set up the frontend and backend dependencies, from the root directory, run:

```
$ npm install --prefix client
$ pipenv install
$ pipenv shell
```

In `server/`, run:

```
$ flask db upgrade
$ python seed.py
```

To see how the React application and Flask API are interacting, you can run the Flask application in one terminal by running:

```
$ python app.py
```

Then, **open another terminal** and run React:

```
$ npm start --prefix client
```

Each application will run on its own port on `localhost` :

- React: <http://localhost:4000> ➡ [_\(http://localhost:4000\)](http://localhost:4000)
- Flask: <http://localhost:5555> ➡ [_\(http://localhost:5555\)](http://localhost:5555)

Deliverables

Routes

Your API should have the following routes as well as the associated controller actions that return the appropriate JSON data:

Update Route

Making a PATCH request to this route with an object in the body should update one plant, and return the updated plant in the response.

```
PATCH /plants/:id
```

Headers

```
Content-Type: application/json
```

Request Body

```
{  
  "is_in_stock": false  
}
```

Response Body

```
{  
  "id": 1,  
  "name": "Aloe",  
  "image": "./images/aloe.jpg",  
  "price": 11.50,  
  "is_in_stock": false  
}
```

Destroy Route

Making a DELETE request to this route should delete one plant from the database. You should return a response of an empty string with the status code 204 (No Content) to indicate a successful deletion.

```
DELETE /plants/:id
```

Response Body

no content

Resources

- [What RESTful Actually Means](https://codewords.recurse.com/issues/five/what-restful-actually-means) ➞ <https://codewords.recurse.com/issues/five/what-restful-actually-means>
- [Flask-RESTful](https://flask-restful.readthedocs.io/en/latest/) ➞ <https://flask-restful.readthedocs.io/en/latest/>
- [HTTP request methods - Mozilla](https://developer.mozilla.org/en-US/docs/Web/HTTP/Methods) ➞ <https://developer.mozilla.org/en-US/docs/Web/HTTP/Methods>
- [Proxying API Requests in Development - React](https://create-react-app.dev/docs/proxying-api-requests-in-development/) ➞ <https://create-react-app.dev/docs/proxying-api-requests-in-development/>

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